

Customer Behavior Analysis Report

1. Executive Summary

This project analyzes customer behavior, revenue drivers, and discount effectiveness using a combination of Python (EDA), SQL (analysis), and Power BI (visualization). The findings show that the business is heavily driven by a large base of loyal customers, with revenue growth primarily dependent on purchase volume rather than higher spending per transaction.

Revenue is concentrated in key categories, with stable demand across locations and minimal seasonal variation, indicating a consistent and predictable market structure. While customer retention appears strong, the dataset shows signs of skew, limiting the reliability of some behavioral conclusions.

Discount strategies are moderately used but do not significantly contribute to revenue growth. They slightly reduce average order value and do not meaningfully improve customer composition or purchase frequency, suggesting limited long-term effectiveness.

Overall, the business should focus on strengthening its core customer segments, optimizing product strategy, and reducing reliance on broad discounting, while validating key data anomalies to ensure more accurate decision-making.

2. Methodology

2.1 Data Exploration (Python)

- Data cleaning & validation
- Distribution analysis
- Customer behavior exploration
- Feature understanding (frequency, purchases, segments)

2.2 Data Analysis (SQL)

- Aggregation:
 - + Revenue by category / location / customer type
- Segmentation:

- + Customer classification (New / Repeat / Loyal)
- Behavioral metrics:
 - + Purchase frequency
 - + Previous purchases
- Performance comparison:
 - + Discount vs non-discount

2.3 Data Visualization (Power BI)

- Dashboard development:
 - + Overview Dashboard
 - + Customer Behavior Dashboard
 - + Revenue Analysis Dashboard
 - + Discount Impact Dashboard
- Interactive filtering:
 - + Gender
 - + Payment method
 - + Shipping type

3. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features: Customer demographics, transactional data (purchase amount, items, purchase history, and frequency), contextual attributes (location, season, shipping, and payment), and marketing variables like discount usage and subscription status, along with review ratings for customer satisfaction.
- 37 missing data in the review rating column, no duplicate records.

4. Data Exploration with Python.

We began with data preparation using Python, here I used Google Colab to practice.

- **Data Loading:** Loading data using **Pandas**
- **Initial Exploration:** Using `df.info()` to check structure and `df.describe()` for summary statistics

```
[4] df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                 3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases             3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
[5] df.describe()

```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Handling null-values:** After checking for null values via `df.isnull().sum()`, any missing entries in the Review Rating columns are filled with the respective category medians.
- **Replacing Header:** Column names are standardized by converting them to lowercase and replacing spaces with underscores. Additionally, `purchase_amount_usd` is renamed to `purchase_amount` to ensure programmatic consistency and ease of use.
- **Binning Age Group:** Binning the `age` column into 'Young Adult', 'Adult', 'Middle-Aged', and 'Senior' groups creates the `age_group` feature. This addition simplifies customer behavior analysis across demographics, with the first 10 rows displayed for verification.

- **Standardization:** Unique values within `frequency_of_purchases` are first identified to explore categorical diversity and detect potential inconsistencies. A `frequency_mapping` dictionary is then utilized to convert these textual frequencies into numerical day values, resulting in the creation of the `purchase_frequency_days` column for quantitative modeling.
- **Data Inspection:** Initial inspection of `discount_applied` and `promo_code_used` displays the first 10 rows to assess value distribution. A full-column comparison yields `np.True_`, confirming that the features are identical for all entries. The `promo_code_used` are then removed from the dataset.

5. Data Analysis with SQL

- **Gender-Based Revenue Analysis:** This query aggregates total revenue by gender to identify the primary revenue-driving demographic for targeted marketing analysis.

	gender	↑↓ ▾	revenue	↑↓ ▾
1	Male		157890	
2	Female		75191	

- **High-Value Discount User Identification:** This query filters for discounted transactions that exceed the average purchase amount to identify high-spending customers who still utilize promotions.

	customer_id	↑↓ ▾	purchase_amount	↑↓ ▾
1	2		64	
2	3		73	
3	4		90	
4	7		85	
5	9		97	
6	12		68	
7	13		72	
8	16		81	

- **Top-Rated Product Identification:** This query identifies the five highest-rated products by calculating the mean **review_rating** for each item, rounded to two decimal places. The results are ordered descending to highlight the top-performing products for quality

benchmarking.

	item_purchased	↑↓	🔍	avg_rating	↑↓	🔍
1	Gloves			3.86		
2	Sandals			3.84		
3	Boots			3.82		
4	Hat			3.8		
5	T-shirt			3.78		

- **Shipping Method Spending Comparison:** Calculation of the average purchase amount for 'Standard' and 'Express' shipping to compare spending behavior across these delivery methods.

	shipping_type	↑↓	🔍	avg_purchase_amount	↑↓	🔍
1	Express			60		
2	Standard			58		

- **Subscription-Based Revenue and Spending Analysis:** This query compares total revenue, average spend, and customer volume between subscribers and non-subscribers to determine the financial impact of the subscription model.

	subscription_status	↑↓	🔍	revenue	↑↓	🔍	avg_spend	↑↓	🔍	number_of_customers	↑↓	🔍
1	Yes			62645			59			1053		
2	No			170436			59			2847		

- **Top Product Discount Affinity Analysis:** Ranking of the top five items with the highest proportion of discounted purchases to identify products most frequently associated with promotional usage.

	item_purchased	↑↓	🔍	discount_ratio	↑↓	🔍
1	Hat			50		
2	Coat			49		
3	Sneakers			49		
4	Sweater			48		
5	Pants			47		

- Customer Loyalty Segmentation Analysis:

Customers are categorized into three distinct tiers based on the frequency of their historical interactions, as defined by the **previous_purchases** metric:

- + **New Customers:** Individuals with exactly **1** recorded purchase, representing the most recent acquisitions.
- + **Returning Customers:** Individuals with **2 to 10** purchases, representing an emerging group with growing brand familiarity.
- + **Loyal Customers:** Individuals with **more than 10** purchases, representing the core high-retention segment.

	customer_segment	Number_of_customer
1	Return Customer	701
2	New Customers	83
3	Loyal Customer	3116

- **Top 3 Products per Category Identification:** This query utilizes a Common Table Expression (CTE) and the **ROW_NUMBER()** window function to rank items within each category based on order volume. By partitioning the data by **category** and ordering by total orders, it isolates the three most popular products for each segment to highlight regional or category-specific top sellers.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- **Repeat Buyer Subscription Correlation:** Evaluation of subscription adoption among high-frequency purchasers (over 5 previous purchases) to determine the correlation between loyalty and membership enrollment.

	subscription_status	↑↓	🔍	repeat_buyers	↑↓	🔍
1	Yes			958		
2	No			2518		

- **Age Group Revenue Contribution:** This query calculates total revenue for each age segment by aggregating **purchase_amount** and ordering the results by financial impact. This breakdown identifies which age demographic serves as the primary revenue driver for the business.

	age_group	↑↓	🔍	total_amount	↑↓	🔍
1	Young Adult			62143		
2	Middle-Aged			59197		
3	Adult			55978		
4	Senior			55763		

6. Visualization with Power BI

The dashboard is divided into four main pages. The **Overview** provides a quick summary of key metrics. The **Customer Behavior** page analyzes customer segments and purchasing patterns. The **Revenue Analysis** page explores where revenue comes from. The **Discount Impact** page evaluates how discounts affect performance and customer behavior.

Overview:



- **Repeat Purchase Rate:** By the categorized column of customer_type, the rate is calculated by: The total of Repeat Customers and Loyal Customers, divided by the Total Customers.
- **Average Order Value:** The value is calculated by: The total of the Purchased Amount divided by the Total Customers.

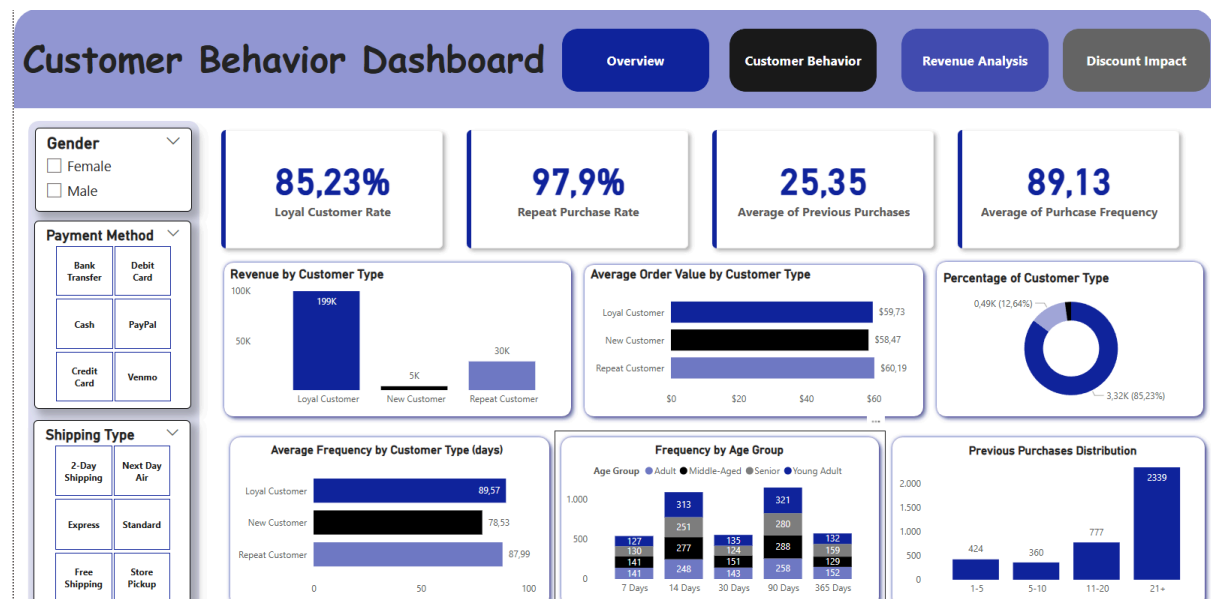
Key Takeaways

- The business generated **\$233K in revenue from ~3.9K customers**, indicating a relatively stable customer base. However, growth potential may be limited without expanding the customer pool, suggesting a need to strengthen acquisition strategies.
- The **repeat purchase rate is extremely high (97.9%)**, which suggests strong retention but is likely influenced by **dataset skew** toward returning customers. This limits the reliability of retention insights and indicates

the need for more balanced data to assess true customer lifecycle behavior.

- Average Order Value (~\$59.76) remains consistent across categories as well as age group, suggesting that customer spending behavior is stable regardless of product type or age group. This implies that revenue growth is more dependent on transaction volume rather than price optimization.
- **Clothing** dominates revenue (~45%), significantly outperforming other categories. This suggests strong customer preference and product-market fit, making it the primary growth driver. The business should prioritize expanding product lines and leveraging cross-selling opportunities from this category.
- **Young Adult and Middle-Aged** customers contribute the majority of revenue and customer volume, making them the core segments. This indicates that marketing and retention efforts should be focused on these groups to maximize impact.
- **Payment method and shipping type show minimal impact**, as their distributions are relatively even across the dataset and do not significantly influence customer behavior or revenue outcomes. This suggests these factors are not key drivers of performance and should not be prioritized in strategic decision-making.

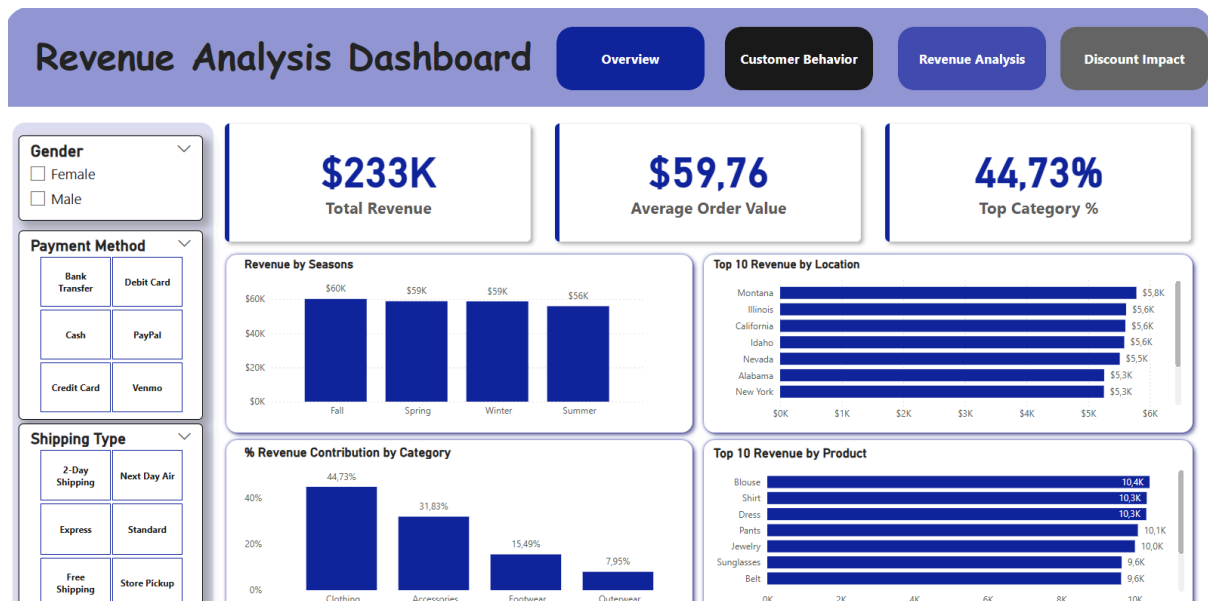
Customer Behavior:



Key Takeaways:

- Loyal customers account for over **85%** of the customer base and generate the majority of revenue (~\$199K). However, their AOV is similar to other segments, indicating that revenue growth is driven by purchase frequency rather than spending per order. This highlights the importance of retention strategies focused on increasing purchase frequency rather than upselling.
- **Loyal and repeat customers exhibit similar purchase frequency, while new customers purchase less frequently**, indicating that customer value is driven more by retention depth than initial engagement. This reinforces the need to focus on converting new customers into repeat buyers through onboarding and early engagement strategies.
- The distribution of previous purchases is heavily concentrated in the **21+ purchases group**, suggesting a strong base of long-term, high-value customers. However, this also indicates potential **dataset skew** and over-reliance on existing customers, highlighting the need to diversify the customer base and reduce dependency on a single segment.
- Purchase frequency shows clustering around **specific intervals (e.g., 14 and 90 days)**, indicating structured buying cycles rather than random behavior. This presents an opportunity to implement **targeted timing strategies** (e.g., reminders, promotions) aligned with these cycles to increase purchase frequency and customer lifetime value.

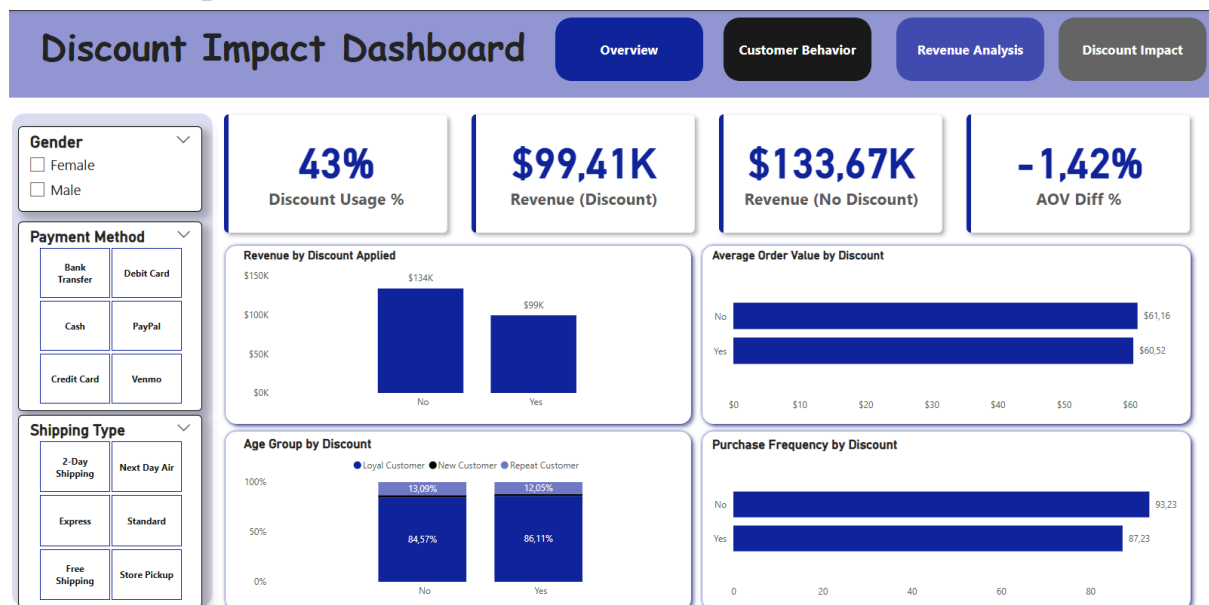
Revenue Analysis:



Key Takeaways:

- Revenue is highly concentrated, with Clothing contributing ~45% of total revenue, indicating a strong dependency on a single category. This suggests high product-market fit but also introduces concentration risk, highlighting the need to diversify revenue streams while further optimizing this core category.
- Revenue is **relatively evenly distributed across locations and top products**, with no extreme outliers. This indicates a balanced market presence and stable demand across regions, suggesting that growth strategies can be scaled consistently rather than focusing on specific geographic hotspots.
- **Seasonal variation is minimal**, indicating that demand remains stable throughout the year. This suggests that revenue performance is not driven by timing factors, and therefore, business growth should focus on structural improvements such as product strategy and customer engagement rather than seasonal campaigns.
- Given these patterns, **growth opportunities lie in optimizing product mix and expanding high-performing categories**, rather than relying on seasonality. This includes strengthening top categories, improving cross-selling strategies, and gradually developing underperforming categories to reduce dependency risk.

Discount Impact:



Key Takeaways:

- **Discount usage is moderate (43%)**, indicating it is a commonly applied tactic but not dominant. This suggests that while discounts are part of the strategy, they are not the primary driver of customer transactions.
- **Non-discounted transactions generate higher total revenue (\$133K vs \$99K)**, indicating that core revenue is driven by full-price purchases. This suggests that the business does not rely heavily on discounting to sustain sales performance.
- **Discounted transactions show a slightly lower AOV (-1.42%)**, indicating that discounts may reduce perceived value or encourage lower spending per order. This suggests a potential margin trade-off without meaningful revenue uplift.
- **Customer composition remains largely unchanged between discounted and non-discounted transactions**, suggesting that discounts do not significantly attract different customer segments. This indicates limited effectiveness in using discounts as a segmentation or acquisition tool.
- **Purchase frequency is slightly lower for discounted transactions**, indicating no strong evidence that discounts improve repeat behavior. However, without time-based or cohort analysis, the long-term impact on engagement cannot be fully validated.

- **No female customers used discounts**, representing a significant anomaly. This may indicate a strong behavioral difference or, more likely, a data quality issue that should be validated before drawing any strategic conclusions.
- Overall, **discount strategies appear to have limited impact on both revenue growth and customer behavior**, suggesting they may function more as short-term incentives rather than long-term value drivers.

7. Recommendation for Business Strategy

a. Prioritized Strategic Focus

Based on the analysis, the business should focus on three key priorities:

1. Optimize discount strategy (immediate impact)
2. Strengthen and leverage the loyal customer base (core driver)
3. Expand high-performing product categories (sustainable growth)

b. Optimize Discount Strategy (Immediate Efficiency Gain)

While discount usage is moderate (43%), it does not significantly improve revenue, customer mix, or purchase frequency. In addition, discounted transactions show a slightly lower AOV (-1.42%), indicating a potential margin trade-off.

- Shift from broad to targeted discounting: Replace blanket discounts with targeted incentives aimed at specific segments, particularly new or low-frequency customers, where impact is more likely.
- Reduce reliance on short-term promotions: Limit general discount campaigns that do not improve long-term engagement or revenue performance.
- Reallocate promotional spend: Redirect resources toward customer acquisition and engagement strategies, especially for underperforming segments and categories.

c. Strengthen Loyal Customer Value (Retention & Expansion)

While the dataset shows an exceptionally high retention rate (97.9%), this is likely influenced by data skew. However, the strong presence of repeat and loyal customers still indicates that retention is a key revenue driver.

- **Implement a value-based loyalty strategy:** Instead of monetary discounts, offer non-price incentives such as early product access, exclusive content, or personalized experiences to maintain engagement without reducing margins.
- **Leverage purchase behavior patterns:** Use identified purchase cycles (e.g., 14-day and 90-day intervals) to trigger timely engagement through reminders or personalized offers.

- **Increase purchase volume through cross-category engagement:** Encourage customers to purchase across multiple categories by offering bundles or cross-category incentives, rather than focusing on single-item discounts.

d. Invest in High-Performing Product Categories (Growth Strategy)

Revenue is highly concentrated in key categories, particularly Clothing (~45%), indicating strong demand and product-market fit.

- **Prioritize incremental investment in core categories:** Expand high-performing product lines and introduce variations within successful categories to maximize return on investment.
- **Target core customer segments:** Focus marketing and product strategies on Young Adult and Middle-Aged customers, who contribute the majority of revenue.
- **Expand through cross-selling and product mix optimization:** Introduce complementary products and bundles to increase basket size and diversify revenue streams.

e. Data Validation and Risk Management

Several findings indicate potential data limitations that should be addressed before making long-term strategic decisions.

- **Validate key anomalies:** Investigate the finding that no female customers used discounts, as this may indicate a data quality issue rather than a true behavioral pattern.
- **Account for dataset bias:** The high proportion of loyal customers may skew insights, limiting the generalizability of retention-related conclusions.
- **Enhance future analysis:** Incorporate time-based or cohort analysis to better evaluate long-term customer behavior and the true impact of promotional strategies.

8. Limitations:

The analysis is subject to several limitations that may impact the reliability and applicability of the findings. The dataset is highly skewed toward repeat and loyal customers, resulting in an unusually high repeat purchase rate that is unlikely to reflect real-world customer behavior. This bias may lead to an overestimation of retention performance and an underestimation of acquisition challenges.

In addition, the absence of female customers using discounts represents a significant anomaly, suggesting either a data quality issue or an unrealistic behavioral pattern. If unvalidated, this could lead to misleading conclusions in segmentation or promotional strategy design.

The dataset also exhibits limited variability, with relatively uniform patterns in average order value, geographic distribution, and seasonal performance. While this simplifies analysis, it reduces the ability to identify meaningful drivers of variation, limiting the depth of insights.

Furthermore, the lack of time-based or cohort-level data prevents evaluation of customer lifecycle behavior and long-term impact of discount strategies. As a result, conclusions regarding retention, engagement, and promotional effectiveness remain static and cannot capture behavioral trends over time.

Finally, as the dataset appears to be synthetic or simplified, it may not fully represent real business complexity. Therefore, findings should be interpreted as directional insights rather than definitive conclusions, and further validation with real-world data is recommended.